Intern Performance Analytics & Insights Dashboard

Sprint 1 – Data Collection & Preparation

1. Overview

The Intern Performance Analytics & Insights Dashboard is a data analytics project designed to evaluate intern performance using structured data. The project focuses on analyzing productivity, attendance, and task completion to generate meaningful insights that can support decision-making.

2. Introduction

In modern organizations, data-driven decision-making plays a crucial role in evaluating employee and intern performance. This project simulates an intern performance tracking system using synthetic data. It involves data creation, preprocessing, analysis, and KPI identification to understand performance trends and prepare for dashboard visualization.

3. Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook
- GitHub

4. Sprint 1 – Day-wise Work

4.1 Day 1 – Dataset Planning

- Understood the project requirements
- Identified key data fields required for performance tracking
- Designed the dataset structure

	A	B	C	D	E	F	G	H
1	Intern_ID	Name	Department	Tasks_Assigned	Tasks_Completec	Attendance	Feedback	Week
2	101	Asha	UI/UX	20	20	75	3.5	Week1
3	102	Ravi	Data Science	21	20	76	3.6	Week2
4	103	Arun	Development	22	20	77	3.7	Week3
5	104	Meena	UI/UX	23	20	78	3.8	Week4
6	105	Kiran	Data Science	24	20	79	3.9	Week5
7	106	Divya	Development	25	25	80	4	Week6
8	107	Surya	UI/UX	20	19	81	4.1	Week7
9	108	Anu	Data Science	21	19	82	4.2	Week8
10	109	Vijay	Development	22	19	83	4.3	Week9
11	110	Priya	UI/UX	23	19	84	4.4	Week10

Technical Terms

Dataset:
A structured collection of data organized in rows and columns.

Column:
A field representing a specific attribute (e.g., Attendance, Tasks).

4.2 Day 2 – Dataset Creation

- Generated a synthetic dataset using Python
- Created data for tasks, attendance, and performance
- Stored the dataset in CSV format

```python
names = ["Aarav", "Diya", "Rahul", "Sneha", "Arjun", "Kiran", "Meena", "Vikram"]

data = {
    "Intern_ID": range(1, 101),
    "Name": np.random.choice(names, 100),    # ✅ NEW COLUMN
    "Department": np.random.choice(["IT", "HR", "Marketing", "Finance"], 100),
    "Tasks_Assigned": np.random.randint(5, 20, 100),
    "Tasks_Completed": np.random.randint(3, 20, 100),
    "Hours_Worked": np.random.randint(80, 200, 100),
    "Attendance": np.random.randint(70, 100, 100),
    "Performance_Score": np.random.randint(50, 100, 100)
}
df["Completion_Rate"] = (df["Tasks_Completed"] / df["Tasks_Assigned"]) * 100
df.head()
```

[4] ✓ 0.0s

	Intern_ID	Name	Department	Tasks_Assigned	Tasks_Completed	Hours_Worked	Attendance	Performance_Score	Completion_Rate
0	1	Meena	Marketing	19	8	174	93	92	42.105263
1	2	Sneha	HR	11	6	133	88	86	54.545455
2	3	Arjun	HR	5	13	137	97	61	260.000000
3	4	Meena	Finance	8	19	146	95	52	237.500000
4	5	Rahul	HR	17	8	183	92	50	47.058824

```
df.to_csv("../dataset/intern_data.csv", index=False)
```

```
df = pd.read_csv("../dataset/intern_data.csv")
df.head()
```

Technical Terms

SyntheticData:
Artificially generated data used when real-world data is unavailable.

Pandas:
A Python library used for data manipulation and analysis.

NumPy:
A library used for numerical operations and generating random values.

DataFrame:
A table-like structure used to store data in Python.

4.3 Day 3 – Data Cleaning

- Checked for missing values
- Handled missing data in the Name column
- Removed duplicate records
- Validated data ranges
- Saved cleaned dataset

```
[13] df.drop_duplicates(inplace=True)

[14] df["Completion_Rate"] = df["Completion_Rate"].clip(0, 100)

[15] df.describe()
```

	Intern_ID	Tasks_Assigned	Tasks_Completed	Hours_Worked	Attendance	Performance_Score	Completion_Rate
count	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000
mean	50.500000	11.830000	10.110000	137.700000	85.890000	75.940000	72.450610
std	29.011492	4.481285	5.489926	34.429404	9.480331	13.764851	28.028544
min	1.000000	5.000000	3.000000	81.000000	70.000000	50.000000	15.789474
25%	25.750000	8.000000	5.000000	105.750000	76.750000	63.000000	50.000000
50%	50.500000	12.000000	9.000000	135.000000	88.000000	75.000000	77.350427
75%	75.250000	16.000000	16.000000	171.250000	95.000000	86.000000	100.000000
max	100.000000	19.000000	19.000000	198.000000	99.000000	99.000000	100.000000

```
df.to_csv("../dataset/intern_data_cleaned.csv", index=False)
```

Technical Terms

DataCleaning:
The process of correcting or removing inaccurate data.

MissingValues:
Empty or null entries in the dataset.

Duplicates:
Repeated rows in the dataset.

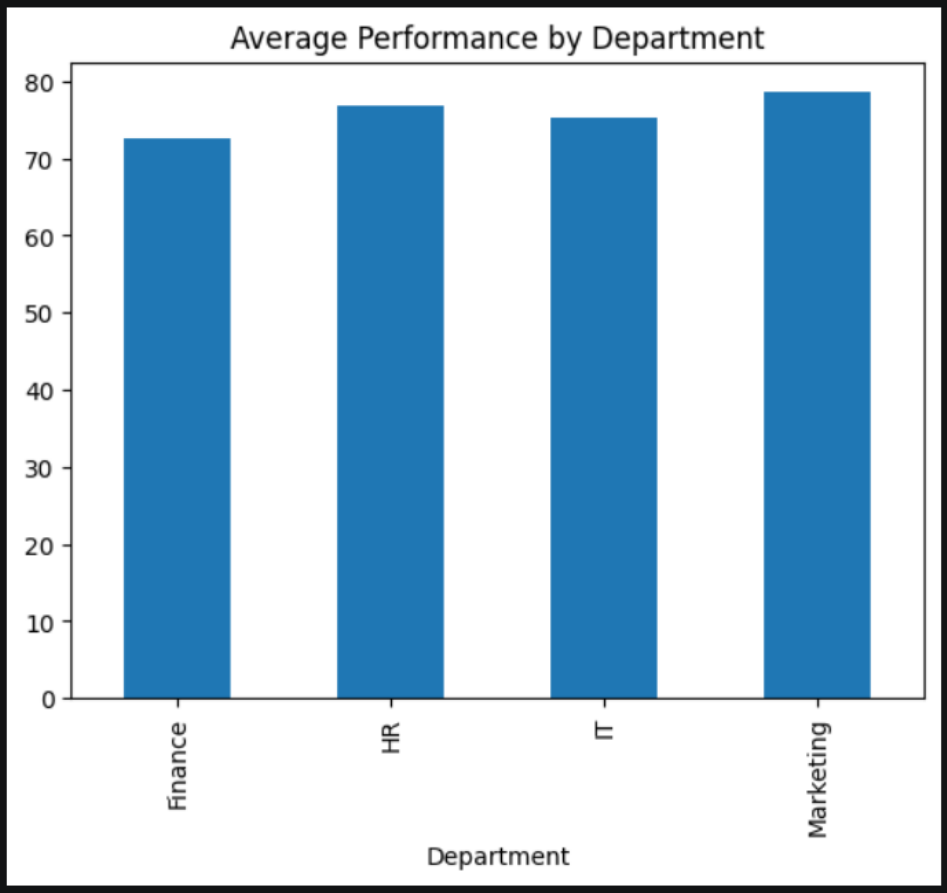
DataValidation:
Ensuring that data values are correct and within expected ranges.

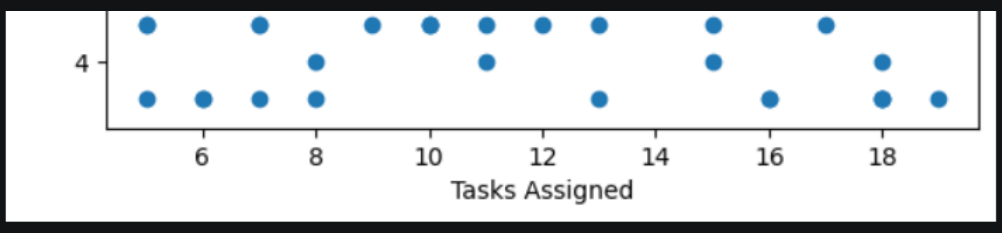
4.4 Day 4 – Exploratory Data Analysis (EDA)

- Created visualizations such as scatter plots, bar charts, and histograms
- Analyzed relationships between variables
- Identified trends and patterns in intern performance

```python
df.groupby("Department")["Performance_Score"].mean().plot(kind="bar")
plt.title("Average Performance by Department")
plt.show()
```

[32]





Technical Terms

EDA(ExploratoryDataAnalysis):
The process of analyzing data using visual and statistical methods.

Visualization:
Graphical representation of data.

ScatterPlot:
A graph used to show the relationship between two variables.

BarChart:
Used to compare values across categories.

Histogram:
Shows the distribution of data.

Correlation:
The relationship between two variables.

4.5 Day 5 – KPI Identification & Dashboard Planning

- Identified key performance indicators (KPIs)
- Calculated performance metrics
- Planned the dashboard structure

```python
df.groupby("Department")[["Performance_Score", "Completion_Rate"]].mean()
```

Department	Performance_Score	Completion_Rate
Finance	72.541667	74.921547
HR	76.850000	67.337656
IT	75.346154	72.503800
Marketing	78.566667	73.836399

```python
import pandas as pd

df = pd.read_csv("../dataset/intern_data_cleaned.csv")
df.head()
```

	Intern_ID	Name	Department	Tasks_Assigned	Tasks_Completed	Hours_Worked	Attendance	Performance_Score	Completion_Rate
0	1	Meena	Marketing	19	8	174	93	92	42.105263
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```python
total_tasks = df["Tasks_Assigned"].sum()
completed_tasks = df["Tasks_Completed"].sum()
avg_completion = df["Completion_Rate"].mean()
avg_performance = df["Performance_Score"].mean()
avg_attendance = df["Attendance"].mean()

print("Total Tasks Assigned:", total_tasks)
print("Total Tasks Completed:", completed_tasks)
print("Average Completion Rate:", avg_completion)
print("Average Performance Score:", avg_performance)
print("Average Attendance:", avg_attendance)
```

```
Total Tasks Assigned: 1183
Total Tasks Completed: 1011
Average Completion Rate: 72.45061020420928
Average Performance Score: 75.94
Average Attendance: 85.89
```

Technical Terms

KPI(KeyPerformanceIndicator):
A measurable value used to evaluate performance.

CompletionRate:
Percentage of tasks completed out of assigned tasks.

PerformanceScore:
A metric used to evaluate overall intern performance.

5. Key Insights

- Intern performance varies across individuals
- Some departments perform better than others
- Attendance has a direct impact on performance
- Completion Rate is a key indicator of productivity

6. Conclusion

This project successfully demonstrates the complete data analysis workflow, including dataset creation, cleaning, and exploratory analysis. By identifying patterns and defining KPIs, the project provides valuable insights into intern performance and lays the foundation for building an interactive dashboard.

7. Future Work (Next Steps)

- Develop an interactive dashboard using Streamlit
- Display KPIs and performance insights visually
- Add filtering options for better analysis
- Integrate real-world data
- Implement role-based access for users

8. Additional Enhancements

- Add predictive analysis using machine learning
- Automate report generation
- Integrate with a database system
- Deploy the dashboard online

9. References

- Python Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib & Seaborn Documentation